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A hybrid genetic algorithm and tabu search for a multi-objective dynamic job shop scheduling problem

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Abstract

In most real manufacturing environments, schedules are usually inevitable with the

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of the problem. A hybrid policy is selected to deal with the dynamic feature of the problem. Two objectives, which are the schedule efficiency and the schedule stability, are considered simultaneously to improve the robustness and the performance of the schedule system. Numerical experiments have been designed to test and evaluate the performance of the proposed method. This proposed method has been compared with some common dispatching rules and meta-heuristic algorithms that have been widely used in the literature. The experimental results illustrate that the proposed method is very effective in various shop-floor conditions.

Keywords:[Dynamic job shop scheduling problem](#)[multi-objective method](#)[hybrid algorithm](#)[schedule efficiency](#)[schedule stability](#)[< Previous article](#)[View issue table of contents](#)[Next article >](#)

1. Introduction

The job shop scheduling problem (JSP) is the process of allocating and timing resource usage in manufacturing system to complete jobs over time according to some desired criteria. It has attracted many studies and engineers because it still exists in most manufacturing systems in various forms (Adibi, Zandieh, and Amiri [2010](#)). Even simplified problems (with deterministic and static assumptions) are NP-hard or analytically intractable (Goren and Sabuncuoglu [2010](#)). A previously feasible schedule in deterministic manufacturing environment may become infeasible when real-time events, such as random job arrivals and machine breakdowns, occur. Taking into account these real-time events, the JSP is termed the dynamic JSP, which is very important and complicated for the successful implementation of real-world scheduling systems (Ouelhadj and Petrovic [2009](#)).

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information. Ouelhadj and Petrovic ([2009](#)) defined the problem and provided a review of the state of the art of currently developing research. They also discussed and compared the scheduling techniques, rescheduling policies, and rescheduling strategies in detail. Recently, a great deal of effort has been spent on developing methods, such as how to reschedule and when to reschedule, to cope with the unexpected disruptions in the real manufacturing scheduling (Ouelhadj and Petrovic [2009](#); Aytug et al. [2005](#)). Moreover, many studies also try to propose a mathematical model or a simulation model to simplify the problem.

The performance measure also plays an important role in addressing the dynamic features in the real manufacturing system. Sabuncuoglu and Goren ([2009](#)) pointed out that developing a bi-criteria approach that considered both the stability and the robustness measures simultaneously warranted further research. Zendieh and Adibi (2010) took the mean flow time as the performance measure. Renna (2010) investigated five performance measures which contain throughput time, throughput, work in process, machine average utilisation and tardiness. In general, most published works which considered the schedule efficiency and schedule stability simultaneously in the dynamic JSP, have not been studied very well.

This paper tries to improve the schedule efficiency and maintain the schedule stability using a method that uses a hybrid genetic algorithm and tabu search with a multi-objective performance measure for dynamic JSP. Moreover, real-time events are difficult to express and take into account using the mathematical model. A simulator is also proposed to tackle the complexity of the problem. The remainder of this paper is organised as follows. Section 2 gives a literature review. The multi-objective dynamic JSP is defined in Section 3. In Section 4, the proposed method is presented. In Section 5, the experimental design and results are discussed. Finally, Section 6 gives the conclusions and future works.

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dynamic JSP with the processing time that was a random variable. Subsequently, a simple heuristic dispatching rule, called the shift from standard rules, was designed by Pierreval and Mebarki ([1997](#)). In the same year, Kouiss, Pierreval, and Mebarki ([1997](#)) first proposed a scheduling strategy based on a multi-agent architecture for dynamic JSP.

Dispatching rules have been widely used in dynamic JSP. Rajendran and Holthaus ([1999](#)) proposed three dispatching rules and considered a total of 13 dispatching rules for dynamic JSP. Subramaniam et al. ([2000](#)) proposed three machine-selection rules. Dominic, Kaliyamoorthy, and Kumar ([2004](#)) provided several efficient dispatching rules for dynamic JSP by combining different dispatching rules. Malve and Uzsoy ([2007](#)) considered a family of iterative improvement heuristics to solve the problem of minimising maximum lateness on parallel identical batch-processing machines with dynamic job arrivals. Alpay and Yuzugullu ([2009](#)) pointed out that due-date assignment model was very successful for improving the missed due-date performance, and the developed dispatching rule was also very successful in meeting the assigned due dates. In order to construct effective scheduling rules for dynamic single-machine scheduling problems, Nie et al. ([2010](#)) proposed the gene-expression programming-based scheduling-rules constructor.

Though many studies discussed rescheduling, no standard definitions or classification of the strategies, policies, and methods have been presented in the rescheduling literature. Vieira, Herrmann, and Lin ([2003](#)) and Ouelhadj and Petrovic ([2009](#)) presented definitions appropriate for most applications of dynamic manufacturing systems. Sabuncuoglu and Kizilisik, [2003](#) proposed several reactive scheduling policies. They found that the full scheduling scheme generally performed better than the partial scheduling. Liu, Ong, and Ng ([2005](#)) presented a framework to model the dynamic JSP as a static group-shop-type scheduling problem, and applied a proposed meta-heuristic for solving the static JSP for a number of dynamic JSP benchmark problems. Sabuncuoglu and Goren ([2009](#)) discussed the major issues involved in

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proposed the combination of auctions with Lagrangian relaxation. Xiang and Lee (2008) built an efficient agent-based dynamic scheduling for real-world manufacturing systems with various products. Renna (2010) created a pheromone-based approach which was carried out by a multi-agent architecture.

Ouelhadj and Petrovic (2009) pointed out that few research works had addressed the use of meta-heuristics in dynamic scheduling. The GA scheduling approach produced better scheduling performance than several common dispatching rules (Chryssolouris and Subramaniam 2001). Zhou, Lee, and Nee (2008) and Zhou, Nee, and Lee (2009) studied ant colony optimisation in the area of dynamic JSP. Adibi, Zandieh, and Amiri (2010) and Zandieh and Adibi (2010) presented a variable neighbourhood search (VNS) method for a multi-objective dynamic JSP. Goren and Sabuncuoglu (2010) showed that the beam search heuristic was capable of generating robust schedules with little average deviation from the optimal objective function value, and it performed significantly better than a number of heuristics available in the literature.

The schedule efficiency implies the high machine utilisation. It can improve the shop efficiency, but the schedule, which only considers the schedule efficiency, may deviate significantly from the original schedule. It can seriously impact other planning activities which are based on the original schedule and may lead to poor performance. In summary, there are many studies that investigate the dynamic JSP. However, the dynamic JSP with a multi-objective, such as the schedule efficiency and the schedule stability simultaneously, has not been studied to any great extent. Moreover, to our knowledge, real-time events are difficult to express and take into account using the mathematical model. Research on the dynamic JSP using classic performance measures such as makespan or tardiness to construct a new schedule may induce instability or a very undesirable effect in shop-floor control (Rangsaritratsamee, Ferrell Jr., and Kurz 2004). Therefore, it is very important to study the dynamic JSP by considering the schedule efficiency and the schedule stability simultaneously. The goal of this paper is to improve the schedule efficiency and

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3 Multi-objective dynamic job shop scheduling

The dynamic JSP is subject to the following technological constraints and assumptions. (1) Each machine can perform only one operation of any job at a time. (2) An operation of a job can be performed by only one machine at a time. (3) All machines are available at time 0. (4) Once an operation has been processed on a machine, it must not be interrupted except for machine breakdown. If an operation is interrupted, the remaining processing time is equal to the total processing time minus the completed processing time. (5) An operation of a job cannot be performed until its preceding operations have been completed. (6) There is no flexible routing for each job. (7) The operation processing time and the number of operable machines are known in advance.

In the practical manufacture system, there are many real-time events, such as random job arrivals and machine breakdowns. In the following section, the effects of the dynamic factors are discussed in detail:

1. Jobs arrive at the system dynamically over time. In dynamic job shops, the distribution of the job arrivals process closely follows a Poisson distribution. Hence, the time between job arrivals closely follows an exponential distribution (Kutanoglu and Sabuncuoglu [2001](#); Rangsaritratsamee, Ferrell Jr., and Kurz [2004](#); Vinod and Sridharan, [2009](#)). When rescheduling is triggered, there are four types of job sets: finished job set, job set being processed, unprocessed job set, and new job set. When the new jobs arrive, the jobs are pushed into the new job set.
2. Machine breakdowns and repairs: the times between two machine failures and the repair time are assumed to follow an exponential distribution. The mean time between failure (*MTBF*) and the mean time to repair (*MTTR*) are two parameters related to machine breakdown (Zandieh and Adibi [2010](#)).

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rescheduling policy is presented for continuous processing in a dynamic environment. In the hybrid rescheduling policy, schedules are generated at regular schedule frequencies and also when a key real-time event appears. A key machine is selected randomly at the beginning of the simulation. The key real-time event occurs when this machine breaks down. A rescheduling will be triggered when the key machine breaks down. A right-shift rescheduling strategy is used when other machines break down. The complete rescheduling strategy, which regenerates a new schedule from scratch, is used at each rescheduling point.

The complete rescheduling strategy may result in instability and lack of continuity in detailed plant schedules, leading to additional production costs attributable to what has been termed shop-floor nervousness (Ouelhadj and Petrovic [2009](#)). This paper attempts to keep the balance between the schedule stability and the schedule efficiency. A multi-objective performance measure is applied as the objective function to construct schedules. The schedule efficiency and the schedule stability are considered in evaluating the solutions. The schedule efficiency factor is evaluated by makespan ($C_{\max}$). A smaller $C_{\max}$ implies a higher machine utilisation (Kashan, Karimi, and Jolai [2010](#)). Now, the makespan traditionally is defined as the total time that is required to process a group of jobs. To fit the dynamic scheduling environment, this definition is modified so that the group of jobs includes all jobs scheduled at each scheduling point. The schedule stability factor is measured by the starting time deviations for all jobs between the new schedule and the original schedule (Ouelhadj and Petrovic [2009](#)). The objective function can be described as:

$$f = \begin{cases} \min & \text{makespan} \\ \min & \text{the starting time deviations} \end{cases} \quad (1)$$

The various weights of these objectives are studied by some researchers and show that weight 5 for makespan and 1 for the starting time deviations are better than the other weights (Rangsaritratamee, Ferrell Jr., and Kurz [2004](#)). The objective function can be formulated as:

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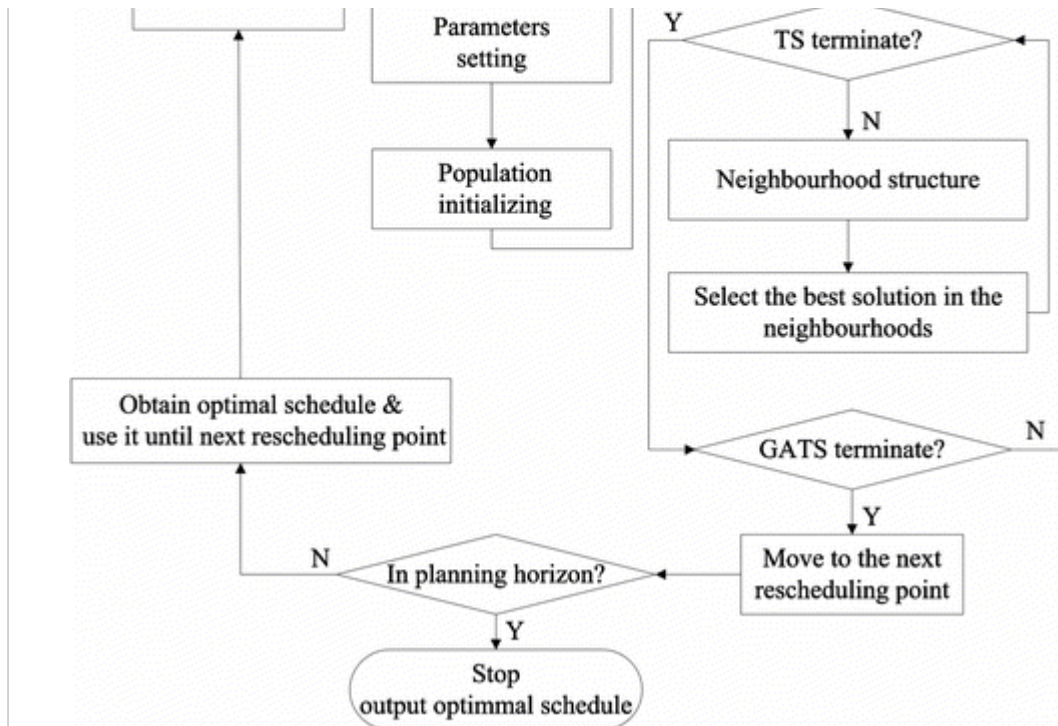
4 Proposed method for dynamic JSP

In this research, a hybrid GA and TS method has been presented by integrating with a simulator to solve the multi-objective dynamic JSP.

4.1 Flow chart for the proposed method

A successful implementation of a scheduling system in a dynamic environment usually involves updating the problem condition, generating and implementing for the production process at each rescheduling point. The general process of the proposed approach is summarised in Figure 1.

Figure 1 Flow chart of the hybrid GA and TS method.

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At the beginning of the scheduling system, all machines are available at time 0. A prediction schedule is predetermined at time 0. Then, the shop floor begins executing the processes as the prediction schedule until the next rescheduling point. Third, if the rescheduling point is smaller than the planning horizon, the reschedule will be triggered. The problem condition, which contains the machine available time, the rescheduling job set, and so on, is updated. The multi-objective genetic algorithm is executed to generate the optimal solutions. The shop floor executes the processes as the optimal schedule. Finally, the above process is repeated until the planning horizon is met. The planning horizon in this work has been defined such that no new job arrives or the number of new jobs reaches a pre-set value.

At each rescheduling point, the machine can be classified into two categories: the machine busy (an operation is being processed on this machine) and the machine idle. In the case of the machine busy, the machine available time will be assumed to

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triggered. The steps of the proposed hybrid GA and TS method are as follows:

Step 1: Set parameters including population size ($popsiz$), maximum iteration of the hybrid GA and TS ($iter_GATS$), maximum iteration of TS ($iter_TS$), crossover probability (p_c), mutation probability (p_m), the length of tabu list (l_t), etc.

Step 2: Randomly generate $popsiz$ population. Decode each individual to obtain the fitness value. Compare them to obtain the initial solution x .

Step 3: Set $n = 1$.

Step 4: Generate new population for the next generations. Genetic evolution with three operators including selection, crossover, and mutation is applied to create offspring for the next population. Let ch_i , $i = 1, 2, \dots, popsiz$ as the new population. The roulette wheel strategy is used to perform the selection.

Step 5: Let each individual of the new population ch_k , $k = 1, 2, \dots, popsiz$ be the initial solution of TS respectively. Repeat steps 6–8 to improve the quality of each individual until iteration is equal to $popsiz$. Then, go to step 9.

Step 6: Set $j = 1$.

Step 7: Find the neighbourhoods of the current solution of TS. Select the best neighbourhood that is non-tabu or satisfies the aspiration criterion.

Step 8: Update the tabu list and check the termination criterion. If the termination criterion ($j = iter_TS$) is met or all critical blocks contain less than one operation, stop TS and output the optimal solution. The individual ch_k is replaced by the optimal solution. Otherwise, $j = j + 1$ and go back to step 7.

Step 9: Check the termination criterion. If the termination criterion ($n = iter_GATS$) is met, stop the hybrid GA and TS method, and output the optimal solution. Otherwise, $n = n + 1$, and go back to step 4.

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Since job arrivals and machine breakdowns occur randomly, a simulator is required to simulate these situations. The simulator, which contains job arrivals and the machine breakdowns and repair, is used to generate the disturbances. The following notations are used for the simulator.

- The *JEL* (jobs event list) denotes the interval between a pair of adjacent job arrivals. When a job arrives, the *JEL* is assigned again. The *AT* represents the job arrival time. Initialise *AT* with zero.
- The *MEL* (machines event list) denotes the interval between failures for each machine. For each machine, when a machine breakdown event occurs, the *MEL* is assigned again. The *MT* represents the time when a machine breaks down. Initialise *MT* with zero.
- The *MREL* (machines repair event list) denotes the interval to repair for each machine. Assume that every machine is available to repair. For each machine, when a machine repair event occurs, the *MREL* is assigned again. The *MR* represents the time when a machine is repaired. Initialise *MR* with zero.
- The *RP* set represents the rescheduling point set. Initialise *RP* with periodic numbers.

The job arrivals simulator has the following steps:

Step 1: Initialise *JEL* with random numbers in exponential distribution; α corresponds to the mean time between a pair of adjacent job arrivals.

$$JEL[J_i] = \exp_rand(\alpha), \text{ for all new jobs } J_i, i = 1, 2, \dots \quad (3)$$

Step 2: Calculate the arrival time of J_i :

$$AT[J_i] = \begin{cases} JEL[J_i] + AT[J_{i-1}] & i > 1 \\ JEL[J_i] & i = 1 \end{cases} \quad (4)$$

Step 3: Repeat Step 1 and Step 2 until the rescheduling point met.

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corresponding to the mean value, and calculate the machine failure time of each machine.

$$MEL[M_k] = \exp_{rand}(MTBF), \text{ for all machines } M_k, k = 1, 2, \dots, m \quad (5)$$

$$MT[M_k] = MEL[M_k] + MR[M_k], \text{ for all machines } M_k, k = 1, 2, \dots, m \quad (6)$$

Step 2: If M_k is the key machine, a rescheduling is triggered; add $MT[M_k]$ to the RP set. Generate an exponential random number where $MTTR$ corresponds to the mean value, and calculate the machine-repair time for each machine.

$$MREL[M_k] = \exp_{rand}(MTTR), \text{ for all machines } M_k, k = 1, 2, \dots, m \quad (7)$$

$$MR[M_k] = MREL[M_k] + MT[M_k], \text{ for all machines } M_k, k = 1, 2, \dots, m \quad (8)$$

Step 3: Repeat Step 1 and Step 2 until the rescheduling point is met.

According to the simulator with respect to the job arrivals simulator and the machine breakdowns and repairs simulator, all disturbances between two adjacent rescheduling points can be obtained.

4.3 Hybrid GA and TS for dynamic JSP

The hybrid GA and TS will be performed to generate an optimal schedule for continuous processing in a dynamic environment at each rescheduling point. It contains chromosome representation and decoding, crossover operator, mutation operator, neighbourhood structure, tabu list and move selection, and termination criterion.

4.3.1 Chromosome representation and decoding

An inappropriate encoding may lead to infeasible solutions in generations after performing crossover and mutation. Therefore, the chromosome design tries to avoid generating infeasible solutions after the crossover and the mutation operations. The operation-based representation method ensures that any permutation of the

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operation of the third job to be processed first on the corresponding machine. Then, the second gene 1 represents the first operation of the first job, and so on. Therefore, the chromosome [3 1 2 2 1 3 1 2 1] is denoted as [O_{31} O_{11} O_{21} O_{22} O_{12} O_{32} O_{13} O_{23} O_{14}] where O_{ij} denotes the j th operation of the i th job. In this research, the operation-based representation method is used to encode a schedule.

Because of jobs arriving randomly, the rescheduling job set contains an unprocessed job set and a new job set at each rescheduling point. A matrix ϑ is selected to represent the rescheduling job set, where O_{ij} denotes the j th operation of job i . j comprises the first unprocessed operations of job i and the maximum number of operations assigned to a resource. For example, when a rescheduling is triggered, the unprocessed job set = { O_{12} O_{13} O_{23} }. The new job set = { O_{31} O_{32} O_{33} }. The rescheduling job set matrix is defined as follows:

$$\vartheta = \begin{pmatrix} O_{12} & O_{13} & \\ O_{23} & & \\ O_{31} & O_{32} & O_{33} \end{pmatrix} \quad (9)$$

Therefore, each job i appears in the chromosome exactly n_i times. For instance, a chromosome [2, 1, 3, 3, 1, 3] is given. Thus, according to the operation-based representation method, the chromosome can be interpreted as [O23 O12 O31 O32 O13 O33].

In the above representation, each possible chromosome always represents a feasible operation sequence. It has been verified and denoted in a Venn diagram in Pinedo (2002) that the active schedule contains the optimal schedule. Therefore, only the active schedule is considered in the decoding approach to reduce the search space.

4.3.2 Crossover operator

Crossover operator determines which parents will have children and tends to increase the quality of the populations. Zhang, Rao, and Li (2008) proposed a

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Step 1: Randomly choose the set of job numbers $\{1, 2, \dots, n\}$ in one non-empty exclusive subset J_1 .

Step 2: Copy those numbers in J_1 from parent1 to child 1 and from parent 2 to child 2, preserving their locus.

Step 3: Copy those numbers in J_1 , which are not copied at Step 2, from parent 2 to child 1 and from parent 1 to child 2, preserving their order.

4.3.3 Mutation operator

Mutation is used to make perturbations on chromosomes in order to maintain the diversity of population. In this research, two types of mutation are applied. One mutation flips the substring between two different random positions. The other mutation selects two elements randomly and inserts the back one before the front one or the front one after the back one.

4.3.4 Neighbourhood structure

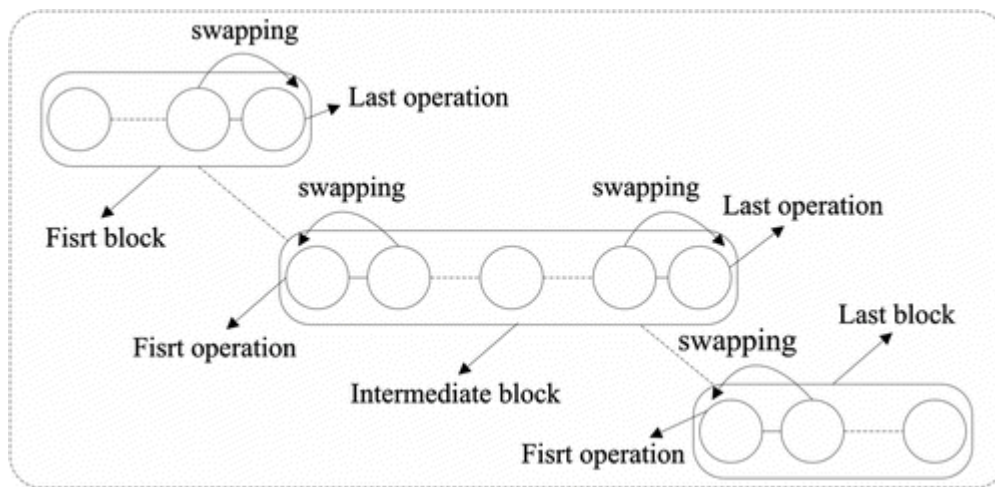
A neighbourhood structure is a mechanism that can obtain a new set of neighbour solutions by applying a small perturbation to a given solution. Neighbourhood structure is directly effective on the efficiency of TS algorithm (Zhang et al. [2007](#)). In this research, the neighbourhood structure N5 proposed by Nowicki and Smutnicki ([1996](#)) is adopted because it is effective for JSP. Nowicki and Smutnicki considered only a single (arbitrarily selected) critical path. However, in order to extend search space, all critical paths are considered.

The neighbourhood structure N5 is a well-known neighbourhood based on the critical block. The critical block is a maximal sequence of adjacent critical operations that is processed on the same machine. Any operation on the critical path, which is the longest path from the start to the end in the directed graph and its length

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symmetry in the last block we swap only the first two operations. Figure 2 illustrates the neighbourhood of moves.

Figure 2 Neighbourhood of moves.



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4.3.5 Tabu list and move selection

The purpose of the tabu list is to avoid the search process turning back to the solutions visited in the previous steps. The length of the tabu list is fixed. The elements stored in the tabu list are the attributes of moves. When the tabu list is full, the oldest element of the list is replaced by the new element. More precisely, if the swapping (x, y) is the best neighbourhood of one chromosome, the swapping (x, y) is added to the tabu list.

Assume that the neighbourhoods of the current solution are not empty. Otherwise, if all critical blocks contain less than one operation, the TS method terminates. The move method is to select the best neighbourhood, which is non-tabu or satisfies the aspiration criterion which accepts the move providing that the fitness value is better than the current best solution found so far. However, there is a situation that all

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The termination criterion is used to determine whether the proposed method should stop. In this research, if the number of iterations is equal to the maximum iterations (iter_TS), or all critical blocks contain less than one operation, TS terminates. If the number of iterations is equal to the maximum number of iterations (iter_GATS), the hybrid GA and TS method terminates. If there are no job arrivals, the planning horizon is met, and the simulator does not generate disturbances.

5 Experimental design and results

5.1 Experimental design

The proposed method has been performed in C++ language on a computer with an Intel Core 2 Duo CPU 2.0 GHz processor and 2.00 GB of memory. Vinod and Sridharan ([2009](#)) pointed out that most of the studies have used between four and 10 machines to test the performance of the schedule system. The example considered in this work is a job shop system with six machines. Simulation starts with a 6×6 static JSP (FT06) (Gao et al. [2011](#)) with *makespan* = 55 extracted from the literature. In order to illustrate the potential of the proposed method for the multi-objective dynamic JSP, it is compared with some common dispatching rules (Zandieh and Adibi [2010](#); Dominic, Kaliyamoorthy, and Kumar [2004](#)) and meta-heuristic algorithms that have widely been used in the literature. A list of the dispatching rules is as follows: (1) The shortest processing time (SPT) dispatching rule. (2) The longest processing time (LPT) dispatching rule. (3) The most work remaining (MWKR) dispatching rule. (4) The least work remaining (LWKR) dispatching rule. (5) The first in first out (FIFO) dispatching rule. (6) The last in first out (LIFO) dispatching rule. A list of the meta-heuristic algorithms is as follows: (1) GA which does not consider tabu search is performed in this paper. (2) Variable neighbourhood search (VNS), which comes from the literature (Zandieh and Adibi [2010](#)), is given in this paper. The setting of parameters in the VNS

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All machines have the same $MTTR$ and $MTBF$. $Ag = MTTR/(MTBF + MTTR)$ denotes the breakdown level of the shop floor that is the percentage of time the machines have failures. For example, for $Ag = 0.05$ and $MTTR = 5$ time units follows $MTBF = 45$. Thus, on an average of 45 time units, a machine is available and then breaks down with a mean time to repair of 5 time units.

The parameter λ (called the job arrivals rate) is the mean of the Poisson random variable. In this research, it denotes the average number of new jobs occurrence per unit time. For example, $\lambda = 0.25$ means that there is an average of 0.25 new job occurrence per unit time. That is to say, a new job arrives at the shop floor in an average of 4 unit times.

In order to evaluate the performance of the proposed method, there are four different job arrivals rate λ and six different number of job arrivals. Therefore, a total of $4 \times 6 = 24$ experiments are conducted. Since the job arrivals rate has been shown to influence the scheduling performance (Shafaei and Brunn [1999](#)), two levels of job arrivals rate, namely sparse and dense job arrivals, are considered. Three types (small, medium, and large) of job arrivals scale are also considered.

The simulation for each state continues until the number of new jobs that have arrived at the shop floor from 20 reaches 200. The makespan and the starting time deviations during the planning horizon are selected as the performance measure. In order to conduct simulation experiments, we created the same experimental conditions (i.e. the number of job arrivals, the number of machines, schedule frequency, and the machine breakdown level). The schedule frequency equals 4 for all experiments. Table 1 illustrates the experimental conditions.

Table 1. Experimental conditions.

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It is worth mentioning that there are several parameters to be determined in the proposed method. Some parameters on the proposed method are taken as: population size ($popsiz = 100$), maximum iteration of the hybrid GA and TS method ($iter_GATS = 20$), maximum iteration of TS ($iter_TS = 20$), crossover probability ($p_c = 0.9$), mutation probability ($p_m = 0.1$), and the length of tabu list ($l_t = 12$). The proposed method runs 10 times for the considered problem. We report the average values of objectives found by the proposed method and other methods. The experimental results are obtained by all methods in the following table and figures.

5.2.1 Experimental results

5.2.1.1 Sparse job arrivals

Table 2 and Table 3 summarise the results of the experiments with sparse job arrivals. The job arrivals rate λ equals 0.125 and 0.25, respectively. Table 2 compares the performance measure of the proposed method with other methods. Because the high schedule efficiency can enhance the machine utilisation, the makespan corresponding to the performance measure is given in Table 3.

Table 2. Performance measure of all methods with sparse job arrivals.

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Table 3. Makespan of all methods with sparse job arrivals.

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These illustrate that the proposed method performs better among these methods, and it can improve the performance of the schedule system by keeping the schedule stability and the schedule efficiency simultaneously when the job arrivals are sparse.

5.2.1.2 Dense job arrivals

Table 4 and Table 5 summarise the results of the experiments with dense job arrivals. The job arrivals rate λ is equal to 0.5 and 1, respectively. Table 4 compares the performance measure of the proposed method with other methods. The makespan corresponding to the performance measure is given in Table 5.

Table 4. Performance measure of all methods with dense job arrivals.

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Table 5. Makespan of all methods with dense job arrivals.

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Table 4 and Table 5 reveal that the proposed method and the VNS outperform these methods. With the increase in the job arrivals, the proposed method for solving the dynamic job shop problems has a better performance than the VNS. These illustrate that the proposed method effectively keeps the balance between the schedule stability and the schedule efficiency, especially in the large scale of job arrivals. Therefore, the proposed method performs better among these methods. In the case of the sparse job arrivals, the proposed method improves the performance of the schedule system by keeping the schedule stability and worsening the schedule efficiency to a small extent when the job arrivals are dense.

methods. The required CPU time is the mean value of each schedule frequency.

Table 6. CPU time of all method with sparse job arrivals.

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Table 7. CPU time of all methods with dense job arrivals.

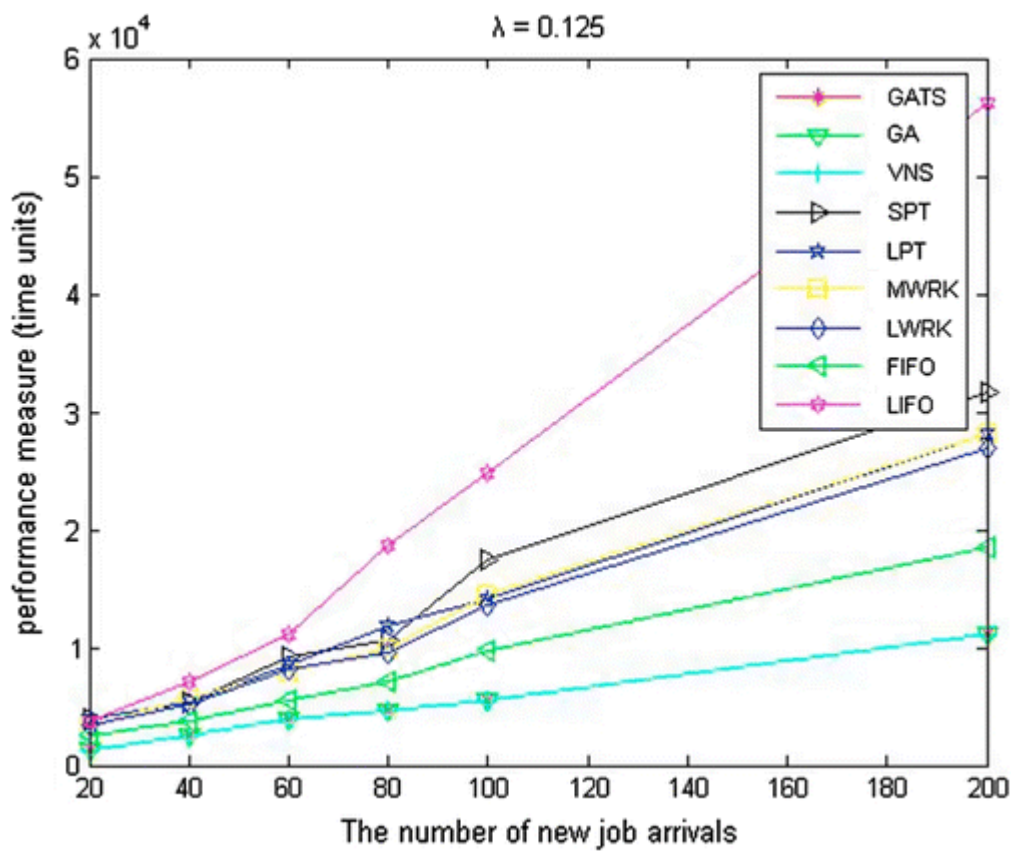
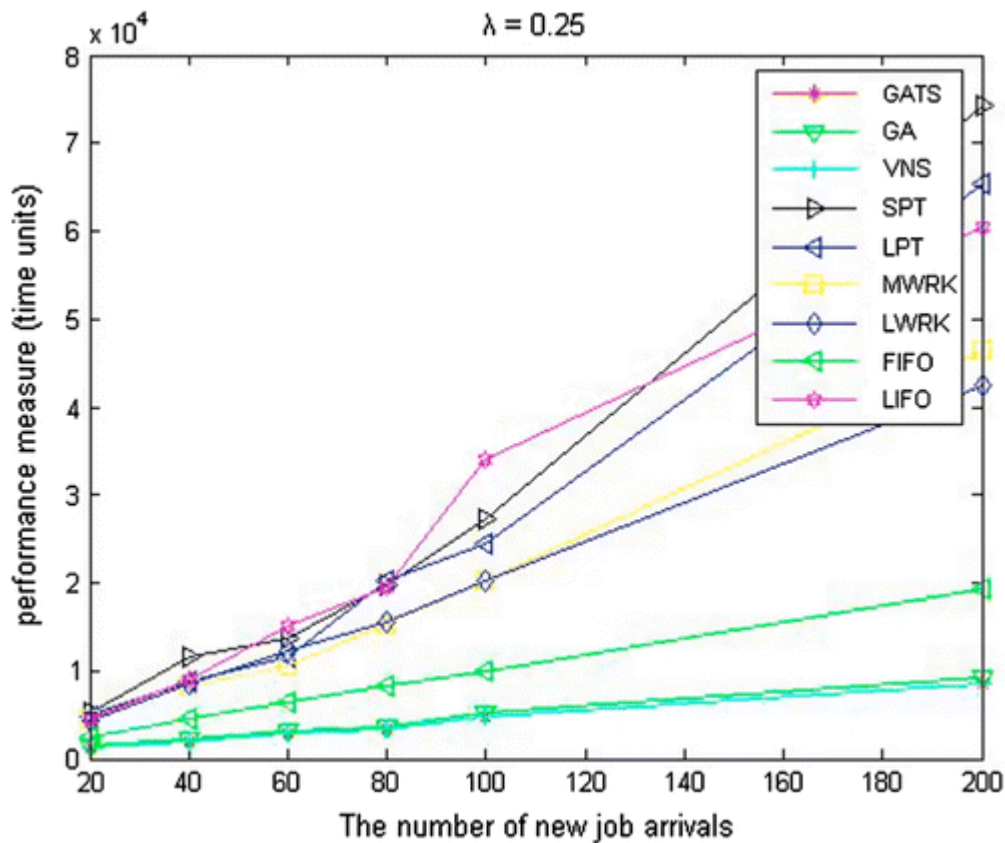
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From Table 6 and Table 7, it can be found that the required CPU times of the dispatching rules are much shorter than for the meta-heuristic algorithms. However, the dispatching rules reduce the performance measure largely, especially in the case of the large and dense job arrivals. The required CPU time is approximate between the proposed method and the VNS. Compared with the dispatching rules, the meta-heuristic algorithms take a longer CPU time, but they can enhance the schedule efficiency and the schedule stability of the schedule system significantly. Otherwise, the required CPU time of the proposed method can be acceptable for real manufacturing systems.

5.2.3 Analysis of their performance in different job arrivals rate, λ

Figure 3 compares the proposed method and other methods in different job arrival rates λ . The x-axis represents the number of job arrivals, and the y-axis represents the performance measure. The job arrivals rate λ of Figure 3(a), Figure 3(b), Figure 3(c), and Figure 3(d) equal 0.125, 0.25, 0.5, and 1, respectively.

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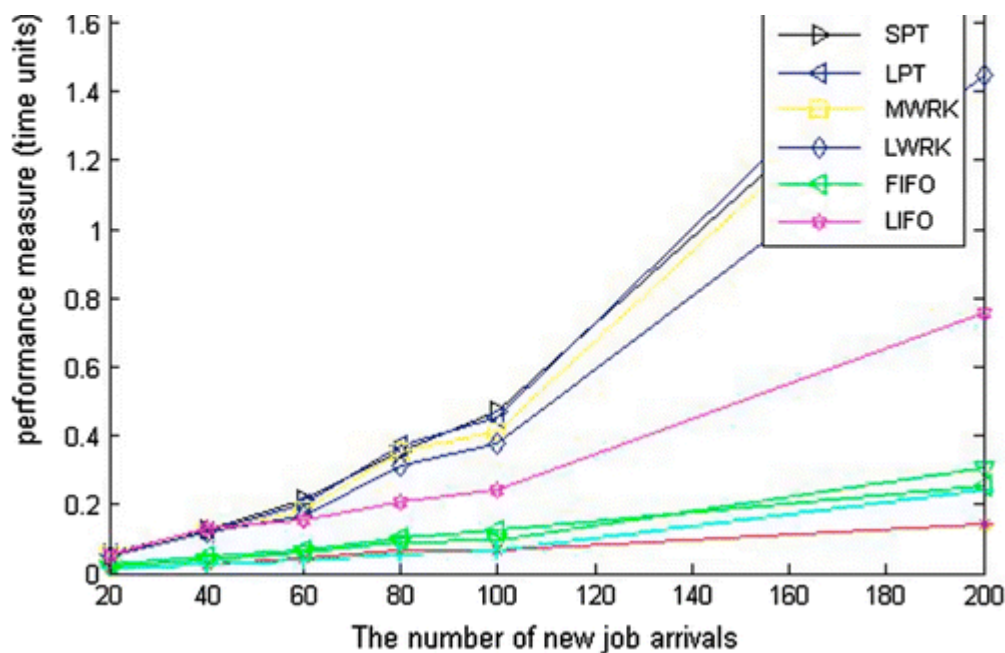


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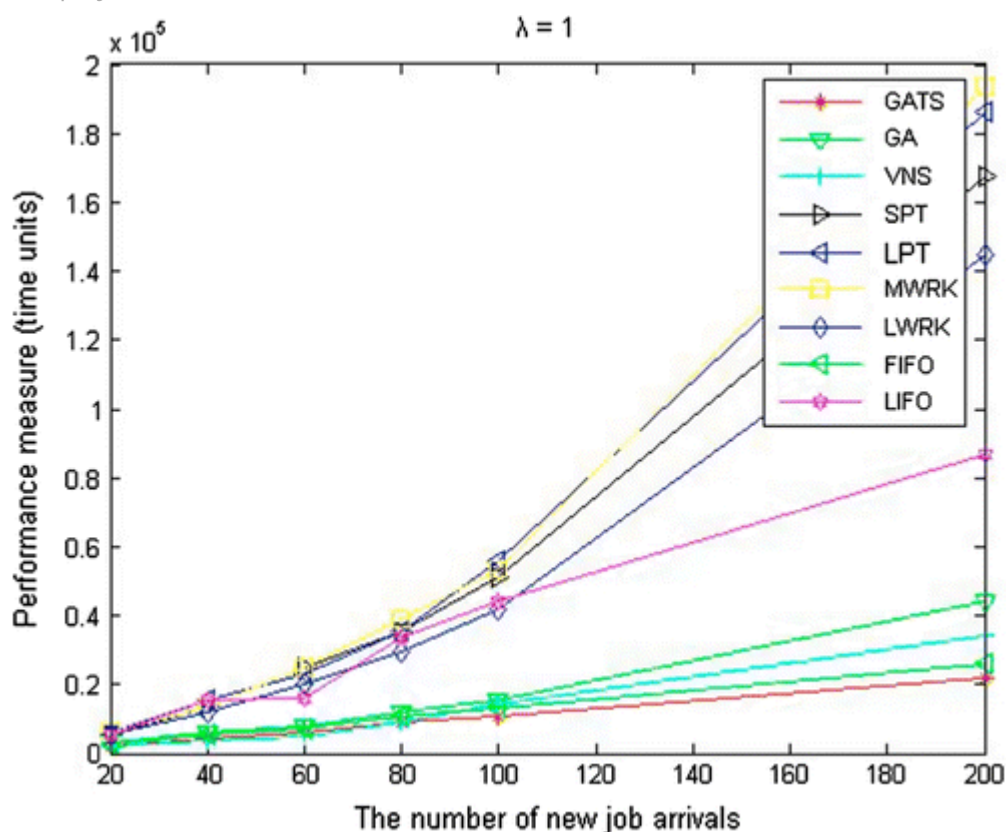
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As indicated in Figure 3, the performance measure increases with increasing number of job arrivals in different job arrivals rate λ . With increasing number of job arrivals,

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stability but the worse schedule efficiency. From Figures 3(a)–(d), the proposed method performs better in the large number of job arrivals under different job arrivals rates. Among these methods, the proposed method shows a good performance in solving the multi-objective dynamic JSP, especially in the large number of job arrivals.

It is also observed that the performance measure of the proposed method and FIFO change approximately linearly with different numbers of new jobs in the four figures. From Figures 3(a)–(d), it also can be found that the angle between the curve of the proposed method and the curve of FIFO becomes smaller when the job arrivals become dense. These illustrate that the proposed method will reduce the schedule efficiency in order to maintain the schedule stability when the job arrivals rate, λ , becomes large.

5.2.4 ANOVA for the performance measure

In order to analyse the performance of the schedule efficiency and the schedule stability with the job arrivals rate λ , the analysis of variance (ANOVA) is performed using the commercial statistical software Microsoft Excel 2010 The ANOVA, given in Table 8, is carried out for the statistical significance test of the factors where factor λ represents the job arrivals rate. In this study, effects are considered significantly if the P-value is less than 0.05.

Table 8. ANOVA for the performance measures.

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As indicated in Table 8, the job arrivals rate has a statistically significant impact on the performance measures of the schedule efficiency and the schedule stability for all

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arrivals rate (0.125, 0.25, 0.5 and 1) form the second factor. Table 9 shows the results of two-factor ANOVA where the main effects and the interaction effect are significant for the performance measures. Effects are considered significant if the P-value is less than 0.05.

Table 9. ANOVA results for two-way analysis.

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As indicated in Table 9, the quantity of new jobs, the levels of job arrivals rate, and the interaction between the quantity of new jobs and the levels of job arrivals rate have a statistically significant impact on the performance measures.

5.2.5 Discussion

Figure 4 shows the main effects of the job arrivals rate on the performance measures under different numbers of the job arrivals. The x-axis represents the job arrivals rate λ , and the y-axis represents the performance measure.

Figure 4 Main effects of the job arrivals rate on the performance measures.

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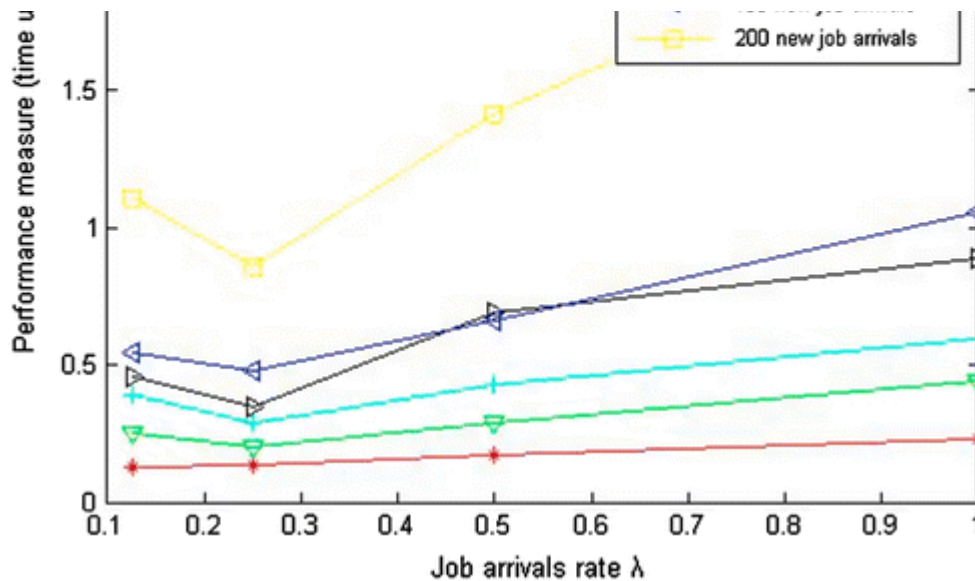


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As indicated in Figure 4, the performance measures are just a similar linear smooth increase in small job arrivals. However, the performance measures changes sharply in large job arrivals. These illustrate that the job arrivals rate λ has a greater impact on the schedule stability and the schedule efficiency, especially in large job arrivals. This conclusion is in accordance with those of the ANOVA on the performance measures.

As indicated in Tables 2–5, the makespan (the schedule efficiency) is affected significantly when the job arrivals is sparse, and the starting time deviations (the schedule stability) is affected significantly when the job arrivals are dense. Figure 4 also reveal that the performance measures are much larger in the job arrivals rate of 1 than others under different number of the job arrivals. A marked inflexion also can be found in the job arrivals rate of 0.25 in Figure 4. That may be because all experiments use the same schedule frequency. When the job arrivals are dense, a large schedule frequency may enhance the schedule stability to improve the performance of the schedule system. However, a small schedule frequency may enhance the schedule efficiency to improve the performance of the schedule system. The influence of the schedule frequency can be discussed in future.

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In this paper, a hybrid algorithm integrated with a simulator is proposed to solve the dynamic job shop scheduling problem with random job arrivals and machine breakdowns. Two performance measures, which contain the schedule efficiency and the schedule stability, are applied in the scheduling process. At any rescheduling point, the simulator generates disturbances for next step. The scheduling scheme is optimised by the hybrid algorithm. The main contributions of this paper are as follows:

- The proposed method has a good performance for solving the dynamic job shop scheduling problem by considering the schedule efficiency and the schedule stability simultaneously in both the sparse/dense job arrivals conditions. In the case of the sparse job arrivals, the proposed method improves the performance of the schedule system by keeping the schedule stability and worsening the schedule efficiency to a small extent when the job arrivals are dense.
- The proposed method takes a longer required CPU time than the dispatching rules. However, it can improve the schedule efficiency and the schedule stability of the schedule system significantly. Otherwise, the required CPU time of the proposed method can be acceptable for a real manufacturing system.
- With the increasing of the job arrivals rate, the angle between the curve of the proposed method and the curve of FIFO becomes small. These illustrate that it is more important to maintain the schedule stability and reduce the 'nervousness' in the dense job arrivals of the schedule system.
- The job arrivals rate λ has a great impact on the schedule stability and the schedule efficiency. The quantity of new jobs and the interaction between the quantity of new jobs and the levels of job arrivals rate also have a statistically significant impact on the performance measures.

The results presented in the paper should be interpreted with reference to the assumptions and experimental conditions described earlier. There are potentially

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reliable, the fault tolerance is an important issue in such systems (Dewan and Joshi [2002](#)). Fault tolerance approaches (Abawajy [2005](#)), such as checkpoint recovery and job replication, and the prediction component (Paulo Leitão [2011](#)) can be applied to the dynamic schedule for future work.

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